

25-11-19

Quantum Mechanics

Wave function :- The probability that a particle will be found at a given place in space at the given instant of time is characterised by the function $\Psi(x, y, z, t)$. It is called the wave function.

- ★ While Ψ itself has no physical interpretation, the square of its absolute magnitude $|\Psi|^2$ (or $\Psi^* \Psi$ if Ψ is complex) evaluated at a particular point at a particular time is proportional to the probability of experimentally finding the particle where at that point.

$\therefore \rho = |\Psi|^2 = \Psi \Psi^*$ where Ψ^* is the complex conjugate of Ψ .

The wave function Ψ can be either real or complex.

- ★ If Ψ is complex. Let $\Psi = A + iB$ where A & B are real functions. The complex conjugate Ψ^* of Ψ is $\Psi^* = A - iB$.

$$\Psi^* \Psi = (A - iB)(A + iB) = A^2 - i^2 B^2 = A^2 + B^2, \text{ since } i^2 = -1.$$

Hence, $\Psi \Psi^*$ is always a positive real quantity.

Probability of finding the particle in the volume element dV is given by $\Psi \Psi^* dV$.

The total probability of finding the particle in the entire space is of course unity. i.e., $\int_{-\infty}^{+\infty} |\Psi|^2 dV = 1 \Rightarrow \int_{-\infty}^{+\infty} \Psi \Psi^* dV = 1$

Any wave function satisfy the above equation is said to be normalized.

Very often Ψ is not a normalized wave function. multiplying Ψ by a constant A , a new wave function will be $A\Psi$ which is also a solution of the wave function. In order that it is a normalized function

$$\int (A\Psi^*) A\Psi dxdydz = 1$$

$$\text{or, } |A|^2 \int \Psi \Psi^* dxdydz = 1$$

$$\text{or, } |A|^2 = \frac{1}{\int \Psi \Psi^* dxdydz}$$

$|A|$ is known as normalizing constant.

Properties of Wave function :-

- 1) The wave function must be normalisable, i.e., $\int_{-\infty}^{+\infty} \Psi \Psi^* = 1$.
- 2) The wave function must be single valued.
- 3) The wave function must be finite everywhere.
- 4) The wave function must be continuous and should be first derivative everywhere.

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Time dependent form of Schrodinger's equation

$$\text{Let, } \Psi = Ae^{-i\omega(t - \frac{x}{v})} \quad \text{--- (1)}$$

$$\therefore \omega = 2\pi\nu \text{ and } v = \frac{c}{\lambda}$$

$$\therefore \Psi = Ae^{-2\pi i(\nu t - \frac{x}{\lambda})} \quad \text{--- (2)}$$

Let E be the total energy and P the momentum of the ~~field~~ particle.

$$\text{Then, } E = h\nu \text{ and } \lambda = \frac{h}{P}$$

$$\therefore \Psi = Ae^{-\left(\frac{2\pi i}{h}\right)(Et - Px)} \quad \text{--- (3)}$$

Differentiating equation (3), twice w.r.t. x , we get,

$$\frac{\partial^2 \Psi}{\partial x^2} = -\frac{4\pi^2 P^2}{h^2} \Psi \quad \text{--- (4)}$$

Differentiating eqⁿ (3) once w.r.t. t , we get,

$$\frac{\partial \Psi}{\partial t} = -\frac{2\pi i E}{h} \Psi \quad \text{--- (5)}$$

$$\text{Since, } E = \frac{p^2}{2m} + V \quad \text{--- (6)}$$

Multiplying both sides of equation (6) by Ψ , we get,

$$\therefore E\Psi = \frac{p^2 \Psi}{2m} + V\Psi \quad \text{--- (7)}$$

$$\text{From eqⁿ (5), } E\Psi = \frac{-h}{2\pi i} \frac{\partial \Psi}{\partial t} \quad \text{--- (8)}$$

$$\text{From eqⁿ (6), } p^2 \Psi = -\frac{h^2}{4\pi^2} \frac{\partial^2 \Psi}{\partial x^2} \quad \text{--- (9)}$$

Substituting the values for $E\Psi$ and $p^2 \Psi$ in eqⁿ (7), we get,

$$\frac{-h}{2\pi i} \frac{\partial \Psi}{\partial t} = \frac{-h^2}{8\pi^2 m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi$$

$$\boxed{\frac{ih}{2\pi} \frac{\partial \Psi}{\partial t} = \frac{-h^2}{8\pi^2 m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi} \quad \text{--- (10)}$$

Eqⁿ (10) is time-dependent form of Schrodinger's Equation.

Schrodinger's equation in a steady state form

$$\text{Let, } \Psi = Ae^{-\left(\frac{2\pi i}{h}\right)(Et - Px)} \\ = Ae^{\left\{-\left(\frac{2\pi i E}{h}\right)t + \left(\frac{2\pi i P}{h}\right)x\right\}}$$

$$\Psi = \Psi_0 e^{\left(\frac{-2\pi i E}{h}\right)t + \left(\frac{2\pi i P}{h}\right)x} \quad \text{--- (11)}$$

$$\text{where, } \Psi_0 = Ae^{\left(\frac{2\pi i P}{h}\right)x}$$

Differentiating eqn (1) w.r.t. t ,

$$\frac{\partial \psi}{\partial t} = -\frac{2\pi i E}{h} \psi_0 e^{-\left(\frac{2\pi i E}{h}\right)t} \quad (2)$$

Differentiating eqn (2), twice w.r.t. x , we get

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{2\pi^2 E}{h^2} \psi_0 e^{-\left(\frac{2\pi i E}{h}\right)t} \quad (3)$$

Substituting these values in time-dependent form of Schrodinger's equations

$$\frac{ik}{2\pi} \left(-\frac{2\pi i E}{h} \right) \psi_0 e^{-\left(\frac{2\pi i E}{h}\right)t} = -\frac{h^2}{8\pi^2 m} \frac{\partial^2 \psi_0}{\partial x^2} e^{-\left(\frac{2\pi i E}{h}\right)t} + V \psi_0 e^{-\left(\frac{2\pi i E}{h}\right)t}$$

$$\psi_0 e^{-\left(\frac{2\pi i E}{h}\right)t} = -\frac{h^2}{8\pi^2 m} \frac{\partial^2 \psi_0}{\partial x^2} + V \psi_0$$

$$\psi_0 e^{-\left(\frac{2\pi i E}{h}\right)t} = -\frac{h^2}{8\pi^2 m} \frac{\partial^2 \psi_0}{\partial x^2} + V \psi_0$$

$$\frac{h^2}{8\pi^2 m} \frac{\partial^2 \psi_0}{\partial x^2} + (E - V) \psi_0 = 0$$

$$\frac{\partial^2 \psi_0}{\partial x^2} + \frac{8\pi^2 m}{h^2} (E - V) \psi_0 = 0 \quad (4)$$

Equation (4) is the steady state Schrodinger's eqn in one dimension.

Steady state Schrodinger's in three dimensions :-

$$\frac{\partial^2 \psi_0}{\partial x^2} + \frac{8\pi^2 m}{h^2} (E - V) \psi_0 = 0$$

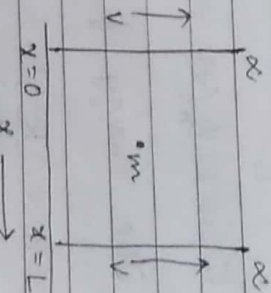
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A particle in a one dimension

The potential energy V of the particle is infinite on both sides of the box. The potential energy V of the particle can be assumed to be zero between $x=0$ and $x=L$.

In terms of boundary conditions imposed by the problems the potential function is

- $V=0$, for $0 < x < L$
- $V=\infty$, for $x \leq 0$
- $V=\infty$, for $x \geq L$



$\psi=0$, for $x \leq 0$
& $\psi=0$ for $x \geq L$
within the box Schrodinger's equation is

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{8\pi^2 m}{h^2} E \psi = 0$$

Putting, $\frac{8\pi^2 m E}{h^2} = k^2$

we have,

$$\frac{\partial^2 \psi}{\partial x^2} + k^2 \psi = 0$$

The general solution of this equation is,
 $\psi = A \sin kx + B \cos kx$ — i)

The boundary condition can be used to evaluate the constant A and B in equation i),

$$\psi = 0 \text{ at } x=0, \therefore B=0$$

$$\text{and, } \psi = 0 \text{ at } x=L,$$

$$\therefore 0 = A \sin kL \text{ if } A=0.$$

Since $A \neq 0 \Rightarrow \psi = 0$, then the probability of finding the particle inside the box is zero.

$$\text{i.e., } kL = n\pi$$

$$k = \frac{n\pi}{L}, n = 0, 1, 2, 3, \dots$$

$$\text{Thus, } \psi_n(x) = A \sin \frac{n\pi x}{L} \text{ — ii)}$$

The energy of the particle,

$$E_n = \frac{k^2 h^2}{8\pi^2 m} = \frac{n^2 h^2 \pi^2}{8\pi^2 m L^2} = \frac{n^2 h^2}{8mL^2} \text{ — iii)}$$

This equation gives the allowed values of energy or the eigen values of the particle. For each values of n , there is an energy level and the

corresponding wave function is given by equation ii). Each value of E_n is called an eigen value and the corresponding ψ_n is called eigen function.

where $n=0$ is not possible because it could make $\psi_n(x) = A \sin nx = 0$ which means that particle is not present at all. The exclusion $n=0$ also means that the particle could not have zero energy. According to classical mechanics all values of energy including 0 is possible. But according to Quantum Mechanics, energy of the particle can not have arbitrary energy but can have discrete value of energy.

Thus, the lowest value of allowed energy is where $n=1$, $E_n = \frac{h^2}{8mL^2}$ — iv)

from equation iii) and iv)

$$E_n = n^2 E_1 \text{ — v)}$$

Putting $n=2, 3, 4, \dots$ in equation iii), we get

$$E_2 = \frac{4h^2}{8mL^2} = 4E_1$$

$$E_3 = 9E_1$$

$$E_4 = 16E_1$$

These are excited energy levels which are not equally spaced.

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Normalized Wave Function

$$\int_0^L |\psi_n(x)|^2 dx = 1$$

$$\int_0^L \psi_n(x) \psi_n^*(x) dx = 1$$

$$\text{or, } A^2 \int_0^L \sin^2 \left(\frac{n\pi x}{L} \right) dx = 1$$

$$\text{or, } A^2 \int_0^L \frac{(1 - \cos 2\pi x/L)}{2} dx = 1$$

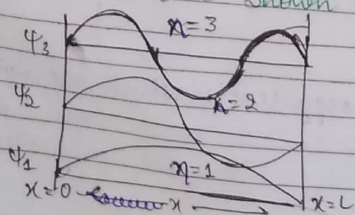
$$\text{or, } \frac{A^2 L}{2} = 1$$

$$\text{or, } A = \sqrt{\frac{2}{L}}$$

Hence, normalized wave function of the particle becomes,

$$\psi_n(x) = \sqrt{\frac{2}{L}} \cdot \sin \frac{n\pi x}{L}$$

Eigen functions ψ_1, ψ_2, ψ_3 corresponding to $n=1, 2, 3$ are shown in figure



Between $x=0$ and $x=L$, there are points where wave function has zero value. These points are called nodes.

Thus, there are $(n+1)$ nodes in wave function ψ_n .
 $\psi_1 = 2$, $\psi_2 = 3$, $\psi_3 = 4$ nodes.

Probability Density :-

Probability of finding the particle between positions x and $x+dx$

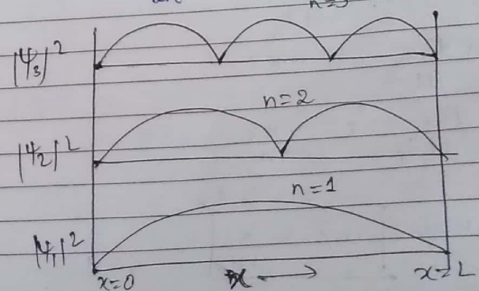
$$P(x) dx = |\psi_n|^2 dx = \frac{2}{L} \sin^2 \frac{n\pi x}{L} dx$$

$$\therefore \text{Probability density } P(x) = \frac{2}{L} \sin^2 \frac{n\pi x}{L}$$

Probability density will be maximum, when,

$$\frac{n\pi x}{L} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \dots$$

$$\therefore x = \frac{L}{2n}, \frac{3L}{2n}, \frac{5L}{2n} \dots$$



The variation of probability densities $(\psi^2/4\pi)^2$ for $n=1, 2, 3$ with 'x' has been shown in figure.

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One-dimensional linear harmonic oscillator

A particle undergoing simple harmonic motion in one dimension is called SHM. harmonic oscillator. In SHM, the restoring force is proportional to displacement.

$$F = -kx$$

where, k is a positive constant called force constant.

According to Newton's second law,

$$F = m \frac{d^2x}{dt^2}$$

$$\text{or } m \frac{d^2x}{dt^2} = -kx$$

$$\text{or } \frac{d^2x}{dt^2} = -\frac{k}{m}x$$

$$\text{where } \boxed{\omega^2 = \frac{k}{m}}$$

$$\text{or } \frac{d^2x}{dt^2} + \omega^2 x = 0 \quad (2)$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$\text{frequency } \nu = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad (3)$$

The potential energy of oscillator is.

$$\begin{aligned} V &= - \int_0^x F dx \\ &= - \int_0^x -kx dx \\ &= k \int_0^x x dx \end{aligned}$$

$$\text{or } V = \frac{kx^2}{2} \quad (4)$$

The one-dimensional schrodinger's time independent form is,

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V) \psi = 0 \quad (5)$$

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} \left(E - \frac{1}{2} kx^2 \right) \psi = 0$$

$|\psi|^2$ or $\psi\psi^* \rightarrow 0$ at $x = \pm\infty$,
we get,

$$2E = (2n+1)\hbar\omega$$

$$\therefore E = (2n+1) \frac{\hbar\omega}{2}$$

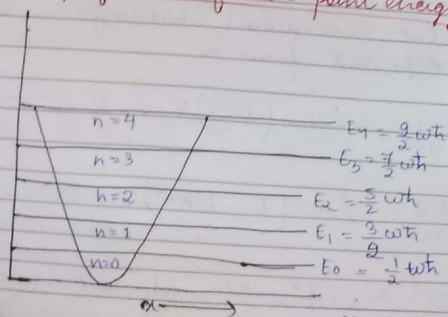
$$E_n = \left(n + \frac{1}{2}\right) \hbar \omega \quad (6)$$

$$n = 0, 1, 2, \dots$$

The allowed integral values of n lead to certain discrete value of energy represented by eq (6) known as eigen value of the harmonic oscillator are equally spaced.

(Moreover eq (6) indicates that the energy the energy level of harmonic oscillator is

Significance of zero point energy:-



For ground state, $x=0$
 $E_0 = \frac{1}{2} \hbar \omega$

Fig 1.

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A comparison with the result $E = n \hbar \omega$ obtained by Old Quantum theory shows that the only difference is that all the equally spaced energy levels are shifted upward by an amount equal to half of the separation of the energy level.
 i.e., $E = \frac{1}{2} \hbar \omega$ = zero point energy.

Thus, it's clear that even in the lowest state, the harmonic oscillator has energy greater than that it could have if it were at rest in its equilibrium position. The existence of zero point energy is in agreement with experiment and it's an important feature of Quantum Mechanics.

The energy levels of harmonic oscillator according to wave mechanics are represented by fig 1 and the corresponding wave functions and their probability densities are respectively shown in fig 2 (a, b, c).

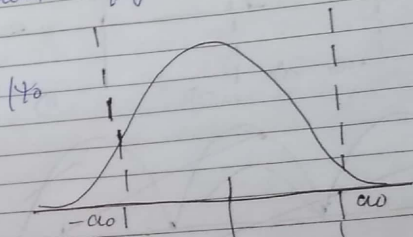


Fig 2(a)

Uncertainty Principle :-

Heisenberg Uncertainty Principle states that the product of uncertainties in determining the position and momentum of the particle is approximately equal to the no. of the order of $\frac{h}{2\pi}$.

$$h = \frac{h}{2\pi}$$

h is planck's constant.

$$\Delta p \cdot \Delta x \approx h \quad \text{--- (1)}$$

where, Δp is the uncertainty in determining the momentum and Δx is the uncertainty in determining the position of the atom.

According to above relation the smaller the value of Δx , more exactly we can determine the position, larger is the value of Δp i.e. less exactly we can determine the momentum and vice-versa. The relation shows that it is impossible to determine

The relation is universal and holds for all the canonically conjugate quantities like position and momentum, energy and time, angular momentum and angle, etc. and whose products are dimension of action of Jsac. (Joule Sec)

Then, if the ΔE is the uncertainty in determining the energy and Δt is the uncertainty in determining the time, through which this determination refers, then we must have this given equations.

$$\Delta p \cdot \Delta x \approx h \quad \text{--- (1)}$$

$$\Delta E \cdot \Delta t \approx h \quad \text{--- (2)}$$

where, Δp = uncertainty in determining of p
 Δx = uncertainty in determination of position
 ΔE = uncertainty in determination of energy
 Δt = uncertainty in determination of time
 ΔJ = uncertainty in determination of angular momentum
 $\Delta \theta$ = uncertainty in determination of angle

Exact Statement of Uncertainty Principle :-

The exact statement of Uncertainty principle is the product of uncertainty's in determining the position and momentum of the particle can never be smaller than the no. of the order of $\frac{1}{2}h$.

$$\therefore \Delta p \cdot \Delta x \geq \frac{1}{2}h \quad \text{--- (4)}$$

$$\Delta E \cdot \Delta t \geq \frac{1}{2}h \quad \text{--- (5)}$$

$$\Delta J \cdot \Delta \theta \geq \frac{1}{2}h \quad \text{--- (6)}$$

Examples:-

- i) Heisenberg γ -ray Microscope
- ii) Diffraction of a beam of electrons by a slit.

Applications:-

- i) The non-existence of electron in the nucleus